Donghyun Kwon

Associate Professor

School of Computer Science and Engineering
Pusan National University

+82 051-510-7338 | kwondh@pusan.ac.kr | https://pnu-csl.github.io/

Education

Seoul National University *Seoul, Korea*
Ph.D in Department of Electrical and Computer Engineering *Mar 2012 - Feb 2019*

- Thesis: Hardware-assisted Isolation for System Framework to Ensure Kernel Integrity
- Advisor: Yunheung Paek
- Distinguished Ph.D. Dissertation Award

Seoul National University *Seoul, Korea*
B.S in Department of Electrical and Computer Engineering *Mar 2008 - Feb 2012*

Professional Experience

Associate Professor *Pusan, South Korea*
School of Computer Science and Engineering, Pusan National University *Mar 2023 - Present*

Assistant Professor *Pusan, South Korea*
School of Computer Science and Engineering, Pusan National University *Mar 2020 - Feb 2023*

Senior Researcher *Daejeon, South Korea*
Electronics and Telecommunications Research Institute *Mar 2019 - Feb 2020*

Publications

** corresponding author*

2026

- **Mind the Gap: In-Process Isolation for Safe Rust in WebAssembly**
  Suhyeon Song, Chaewon Shin, and **Donghyun Kwon***
  European Symposium on Research in Computer Security (ESORICS), 2026
- **XCFI: Comprehensive Control-Flow Integrity for Arm TrustZone-M**
  Yunju Gu, Jaeyeol Park, and **Donghyun Kwon***
  USENIX Security Symposium, 2026
- **TzPy: Python Runtime for Edge Code Deployment on TrustZone**
  Suhyeon Song, Jaeyeol Park, Chaewon Shin, and **Donghyun Kwon***
  ACM Transactions on Embedded Computing Systems (TECS), 2026
- **SERA: Secure Micro XRCE-DDS Establishment with Remote Attestation for micro-ROS**
  Jeonghwan Kang, Seungmin Kang, Euijun Kim, Jaeseon Park, Kyounghwan Kim, and **Donghyun Kwon***
  IEEE Access, 2026
- **WARD: Efficient Memory Protection for WebAssembly on Tiny Embedded Systems**
  Chaewon Shin, Gowon Han, Suhyeon Song, Jinsun Park, Yoon-ho Choi, and **Donghyun Kwon***
  IEEE Access, 2026
- **Beyond Address Spaces: In-process Memory Isolation for RISC-V**
  Seonghwan Park, Hayoung Kang, and **Donghyun Kwon***
  Computers & Security, 2026

2025

- **SMORE: Practical Redzone-Based Stack Memory Error Detection Mechanism for Embedded Systems**
  Jaeyeol Park, Yunju Gu, and **Donghyun Kwon***
  Annual Computer Security Applications Conference (ACSAC), 2025
- **ELK: Effective Lock-and-Key Technique for Temporal Memory Safety on Embedded Devices in ARMv8-M**
  Jeonghwan Kang, Kyounghwan Kim, and **Donghyun Kwon***
  Annual Computer Security Applications Conference (ACSAC), 2025
- **Watch Your Callback: Offline Anomaly Detection using Machine Learning in ROS 2**
  Jeonghwan Kang, Kyounghwan Kim, and **Donghyun Kwon***
  IEEE Access, 2025
- **WasDom: An Efficient Write Protection for Wasm JITed Code With ARM Domain**
  Suhyeon Song, Chaewon Shin, and **Donghyun Kwon***
  IEEE Access, 2025
- **ROSec: Intra-Process Isolation for ROS Composition with Memory Protection Keys**
  Jiwon Seo, Kayondo Martin, Jeonghwan Kang, **Donghyun Kwon***, and Yunheung Paek*
  IEEE Transactions on Automation Science and Engineering (T-ASE), 2025

2024

- **Non-volatile flip-flop with soft error tolerant capability using DICE and C-element**
  Shogo Takahashi, **Donghyun Kwon**, and Kazuteru Namba*
  IEICE Nonlinear Theory and Its Applications (NOLTA), 2024
- **Look Before You Access: Efficient Heap Memory Safety for Embedded Systems on ARMv8-M**
  Jeonghwan Kang, Jaeyeol Park, Jiwon Seo, and **Donghyun Kwon***
  Design Automation Conference (DAC), 2024
- **MECAT: Memory-Safe Smart Contracts in ARM TrustZone**
  Seonghwan Park, Hayoung Kang, Sanghun Han, Jonghee M. Youn*, and **Donghyun Kwon***
  IEEE Access, 2024

2023

- **Enhancing a Lock-and-key Scheme with MTE to Mitigate Use-After-Frees**
  Inyoung Bang, Martin Kayondo, Junseung Yu, **Donghyun Kwon**, Yeongpil Cho*, and Yunheung Paek*
  IEEE Access, 2023
- **metaSafer: A Technique to Detect Heap Metadata Corruption in WebAssembly**
  Suhyeon Song, Seonghwan Park, and **Donghyun Kwon***
  IEEE Access, 2023
- **ZOMETAG: Zone-based Memory Tagging for Fast, Deterministic Detection of Spatial Memory Violations on ARM**
  Jiwon Seo, Junseung You, **Donghyun Kwon**, Yeongpil Cho*, and Yunheung Paek*
  IEEE Transactions on Information Forensics and Security (TIFS), 2023
- **SFITAG: Efficient Software Fault Isolation with Memory Tagging for ARM Kernel Extensions**
  Jiwon Seo, Junseung You, Yungi Cho, Yeongpil Cho, **Donghyun Kwon***, Yunheung Paek*
  ACM ASIA Conference on Computer and Communications Security (ASIACCS), 2023
- **DEMIX: Domain-Enforced Memory Isolation for Embedded System**
  Haeyoung Kim, Harashta Tatimma Larasati, Jonguk Park, Howon Kim, and **Donghyun Kwon***
  Sensors, 2023

2022

- **Exploring Effective Uses of the Tagged Memory for Reducing Bounds Checking Overheads**
  Jiwon Seo, Inyoung Bang, Yungi Cho, Jangseop Shin, Dongil Hwang, **Donghyun Kwon**, Yeongpil Cho, Yunheung Paek
  The Journal of Supercomputing, 2022
- **Efficient CFI Enforcement for Embedded Systems Using ARM TrustZone-M**
  Gisu Yeo, Yeryeong Kim, Suhyeon Song, and **Donghyun Kwon***
  IEEE Access, 2022
- **Bratter: An Instruction Set Extension for Forward Control-Flow Integrity in RISC-V**
  Seonghwan Park, Dongwook Kang, Jeonghwan Kang, and **Donghyun Kwon***
  Sensors, 2022

2021

- **CoMeT: Configurable Tagged Memory Extension**
  Jinjae Lee, Derry Pratama, Minjae Kim, Howon Kim, and **Donghyun Kwon***
  Sensors, 2021
- **A Hardware Platform for Ensuring OS Kernel Integrity on RISC-V**
  Donghyun Kwon, Dongil Hwang, and Yunheung Paek
  Electronics, 2021

2020

- **RIMI: Instruction-level Memory Isolation for Embedded Systems on RISC-V**
  Haeyoung Kim, Jinjae Lee, Derry Pratama, Asep Muhamad Awaludin, Howon Kim, and **Donghyun Kwon***
  International Conference on Computer-Aided Design (ICCAD), 2020
- **PrOS: Light-weight Privatized Secure OSes in ARM TrustZone**
  Donghyun Kwon, Jiwon Seo, Yeongpil Cho, Byoungyoung Lee, and Yunheung Paek
  IEEE Transactions on Mobile Computing (TMC), 2020

2019

- **RiskiM: Toward Complete Kernel Protection with Hardware Support**
  Dongil Hwang, Myunghoon Yang, Seongil Jeon, Younghan Lee, **Donghyun Kwon***, and Yunheung Paek
  Design, Automation and Test in Europe (DATE), 2019
- **Safe and Efficient Implementation of a Security System on ARM using Intra-Level Privilege Separation**
  Donghyun Kwon, Hayoon Yi, Yeongpil Cho, and Yunheung Paek
  ACM Transactions on Privacy and Security (TOPS), 2019
- **CRCount: Pointer Invalidation with Reference Counting to Mitigate Use-After-Free in Legacy C/C++**
  Jangseop Shin, **Donghyun Kwon**, Jiwon Seo, Yeongpil Cho, and Yunheung Paek
  Network and Distributed System Security Symposium (NDSS), 2019
- **uXOM: Efficient eXecute-Only Memory on ARM Cortex-M**
  Donghyun Kwon, Jangseop Shin, Giyeol Kim, Byoungyoung Lee, Yeongpil Cho, and Yunheung Paek
  USENIX Security Symposium, 2019

2018

- **Hypernel: A Hardware-Assisted Framework for Kernel Protection without Nested Paging**
  Donghyun Kwon**, Kuenwhee Oh**, Junmo Park, Seungyong Yang, Yeongpil Cho, Brent Byunghoon Kang, and Yunheung Paek
  Design Automation Conference (DAC), 2018 (** joint first authors)
- **VM-CFI: Control-Flow Integrity for Virtual Machine Kernel Using Intel PT**
  Donghyun Kwon, Jiwon Seo, Sehyun Baek, Giyeol Kim, Sunwoo Ahn, and Yunheung Paek
  International Conference on Computational Science and Its Applications (ICCSA), 2018

2017

- **Instruction-Level Data Isolation for the Kernel on ARM**
  Yeongpil Cho, **Donghyun Kwon**, and Yunheung Paek
  Design Automation Conference (DAC), 2017
- **Optimization Techniques to Enable Execution Offloading for 3D Video Games**
  Donghyun Kwon, Seungjun Yang, Yunheung Paek, and Kwangman Ko
  Multimedia Tools and Applications (MTAP), 2017
- **Dynamic Virtual Address Range Adjustment for Intra-Level Privilege Separation on ARM**
  Yeongpil Cho, **Donghyun Kwon**, Hayoon Yi, and Yunheung Paek
  Network and Distributed System Security Symposium (NDSS), 2017

2016

- **Precise Execution Offloading for Applications with Dynamic Behavior in Mobile Cloud Computing**
  Yongin Kwon, Hayoon Yi, **Donghyun Kwon**, Seungjun Yang, Yeongpil Cho, and Yunheung Paek
  Pervasive and Mobile Computing (PMC), 2016
- **Hardware-Assisted On-Demand Hypervisor Activation for Efficient Security Critical Code Execution on Mobile Devices**
  Yeongpil Cho, Jun-Bum Shin, **Donghyun Kwon**, MyungJoo Ham, Yuna Kim, and Yunheung Paek
  USENIX Annual Technical Conference (ATC), 2016

2015

- **Mantis: Efficient Predictions of Execution Time, Energy Usage, Memory Usage and Network Usage on Smart Mobile Devices**
  Yongin Kwon, Sangmin Lee, Hayoon Yi, **Donghyun Kwon**, Seungjun Yang, Byung-gon Chun, Ling Huang, Petros Maniatis, Mayur Naik, and Yunheung Paek
  IEEE Transactions on Mobile Computing (TMC), 2015

2014

- **Techniques to Minimize State Transfer Costs for Dynamic Execution Offloading in Mobile Cloud Computing**
  Seungjun Yang, **Donghyun Kwon**, Hayoon Yi, Yeongpil Cho, Yongin Kwon, and Yunheung Paek
  IEEE Transactions on Mobile Computing (TMC), 2014

2013

- **Fast Dynamic Execution Offloading for Efficient Mobile Cloud Computing**
  Seungjun Yang, Yongin Kwon, Yeongpil Cho, Hayoon Yi, **Donghyun Kwon**, Jonghee Youn, and Yunheung Paek
  IEEE International Conference on Pervasive Computing and Communications (PerCom), 2013
- **Mantis: Automatic Performance Prediction for Smartphone Applications**
  Yongin Kwon, Sangmin Lee, Hayoon Yi, **Donghyun Kwon**, Seungjun Yang, Byung-Gon Chun, Ling Huang, Petros Maniatis, Mayur Naik, and Yunheung Paek
  USENIX Annual Technical Conference (ATC), 2013

Academic Service

- (Reviewer) ACM Transactions on Privacy and Security (TOPS)
- (Reviewer) ACM Transactions on Architecture and Code Optimization (TACO)
- (Reviewer) IEEE Transactions on Information Forensics & Security (TIFS)
- (Reviewer) IEEE Transactions on Computers (TC)
- (Reviewer) IEEE Access
- (Reviewer) Journal of Supercomputing
- (Associate Editor) Journal of Information Processing Systems (JIPS)

Teaching

Graduate

- **System Vulnerabilities and Countermeasures** — F21, F22, F23, F24, F25
- **Introduction to System and Software Security** — S21, S22, S23, S24, S25, S26

Undergraduate

- **Web Application Programming** — F21, F22, F23
- **Logic Circuit Design & LAB.** — F23, F24
- **Compiler** — F22
- **Computer Architecture** — F24, F25
- **Blockchain** — S24, S25, S26
- **Programming Principles and Practice** — F20, F21
- **Adventure Design** — F20
- **Introduction to Internet and Web** — S20, S21, S22, S23
- **System Software** — S20, S21, S22, S23, S24, S25, S26
- **Discrete Mathematics 1** — S20